

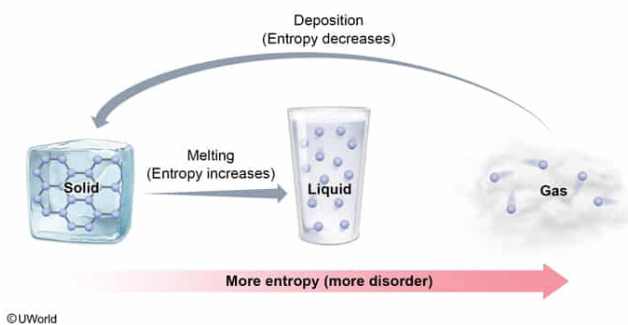
A sample of water vapor undergoes deposition and then melting. What happens to its entropy after each phase change?

- ☐ A. Increased both times. [2%]
- ☐ B. Decreased both times. [2%]
- ☐ C. Increased with deposition, decreased with melting [5%]
- ✓ ☒ D. Decreased with deposition, increased with melting [89%]

Correct

89% answered correctly

Explanation:



Matter can exist in three different states (or phases): solid, liquid, and gas. These phases describe the uniform physical state of a substance's molecules:

- Molecules in a solid are tightly packed in an ordered structure and retain a fixed shape.
- Molecules in a liquid move past each other but stay close together. A liquid retains a fixed volume but assumes the shape of its container.
- Molecules in a gas move past each other and are well separated. A gas assumes the shape and volume of its container.

Entropy is the thermodynamic quantity that measures the **disorder** of a substance (or system). Solids have relatively low entropy because the structure of the molecules is ordered and their motion is more constrained. Conversely, gases have relatively high entropy because the molecules are disordered and their motion is less constrained.

Therefore, entropy changes as matter **transitions** from one phase to another. The sample of water vapor first undergoes deposition and then melting.

Deposition is the phase transition from **gas to solid**, and solids have lower entropy than gases; therefore, **entropy decreases (Choices A and C)**.

Melting is the phase transition from **solid to liquid**, and liquids have higher entropy than solids; therefore, **entropy increases (Choice B)**.

Educational objective:

Matter can transition between solid, liquid, and gas phases. Entropy measures the disorder of a substance. Changing from gas to solid (deposition) decreases entropy, and changing from solid to liquid (melting) increases entropy.

Time spent:0 sec

Subject:

Physics

QID:401352

System:

Thermodynamics